Effectiveness of Digital Cognitive Training Interventions for Dementia: A Systematic Review and Meta-Analysis

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BACKGROUND

- Current pharmacological treatments primarily focus on symptom management and have **limited effectiveness** in halting the progression of the disease.
- Technologies such as smartphone applications and wearable devices provide effective means **for early diagnosis and prevention**.
- Dementia treatment and care represent a high-cost sector within healthcare. Digital cognitive training interventions for dementia provide a **cost-efficient alternative** by reducing hospital visits and enabling care at home.

OBJECTIVES

- The field of digital cognitive training interventions for dementia is rapidly evolving, this research helps **track these developments** and **assess their relevance and impact**.
- This research can serve as a **foundation for interdisciplinary collaboration** such as medicine, pharmacy, computer science by providing a unified overview of the field.
- This research identifies gaps in existing research, **guiding future studies** and focusing efforts on areas with high potential for dementia care.

METHODS

- Study protocol preregistered with the PROSPERO registry (CRD42025631806)

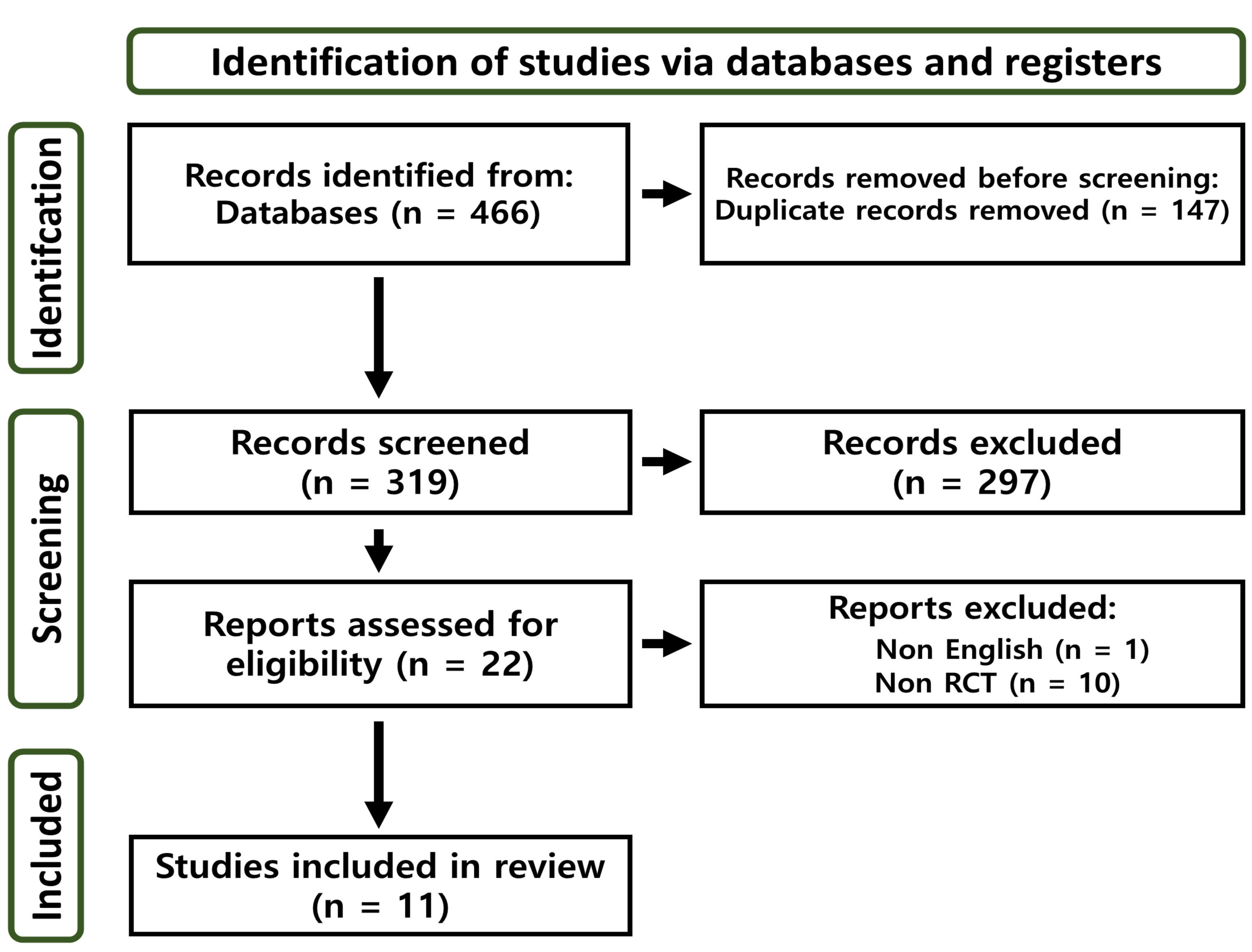

Figure 1. Flow diagram of literature search using the PRISMA model

- A comprehensive literature search was conducted 3 databases(PubMed, Embase, Cochrane) using a search strategy based on PRISMA.
- This search was performed on randomized controlled trial studies published in English Until December 2024.
- All types of dementia patients are included, excluding mild cognitive impairment (MCI) and subjective cognitive decline (SCD) patients.
- Includes cases where patients can perform programs remotely using software, applications, or mobile games.

RESULTS

- The total number of patients who participated in the studies was 1,352, with 636 (M/F : 228/407/1*) in the intervention group and 716 (M/F : 300/416) in the control group.

* Gender: Non-response

- The average age of the intervention group that participated in the studies was 82.67±7.20, while that of the control group was 82.74±7.22.

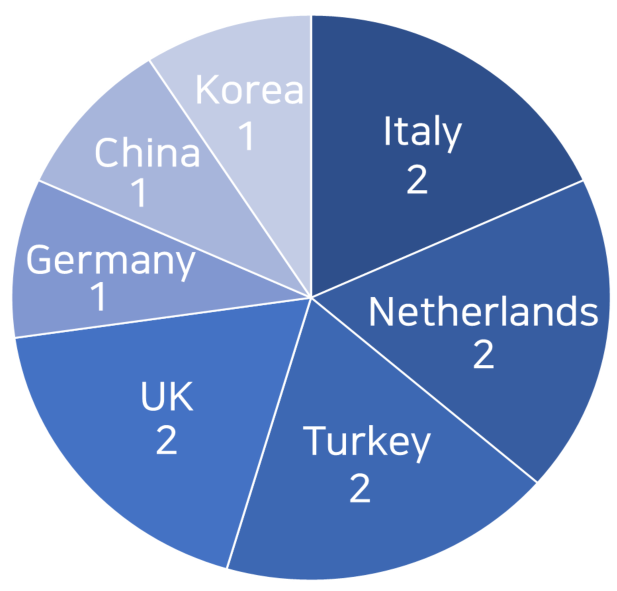

Figure 2-1. First author's nationality in the studies

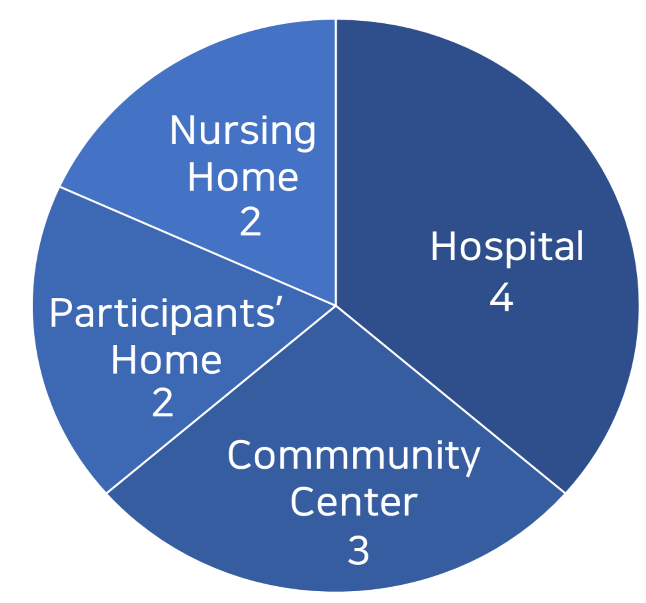

Figure 2-2. Settings in the studies

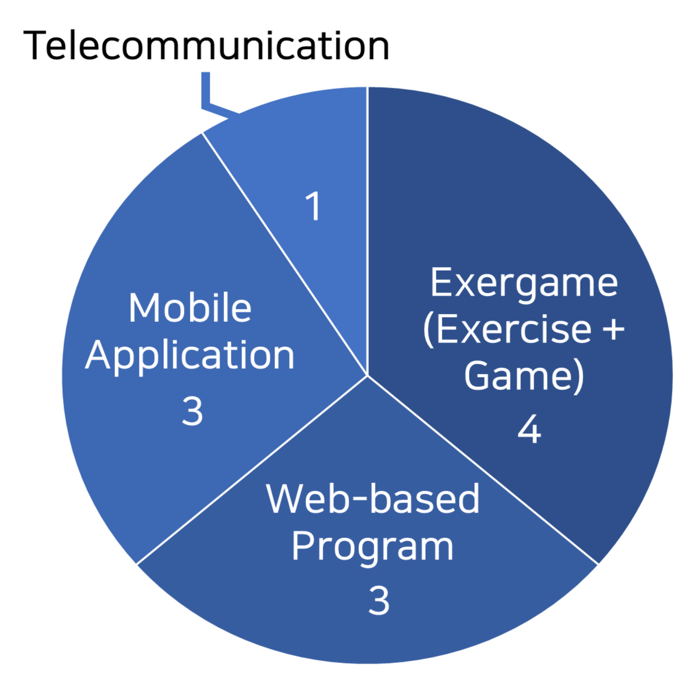

Figure 2-3. Interventions in the studies

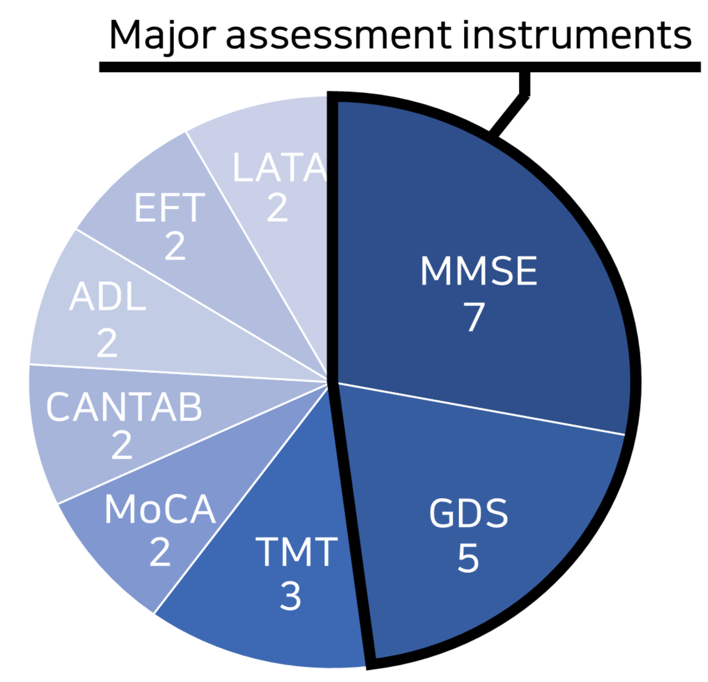

Figure 2-4. Frequency of assessment instrument in the studies

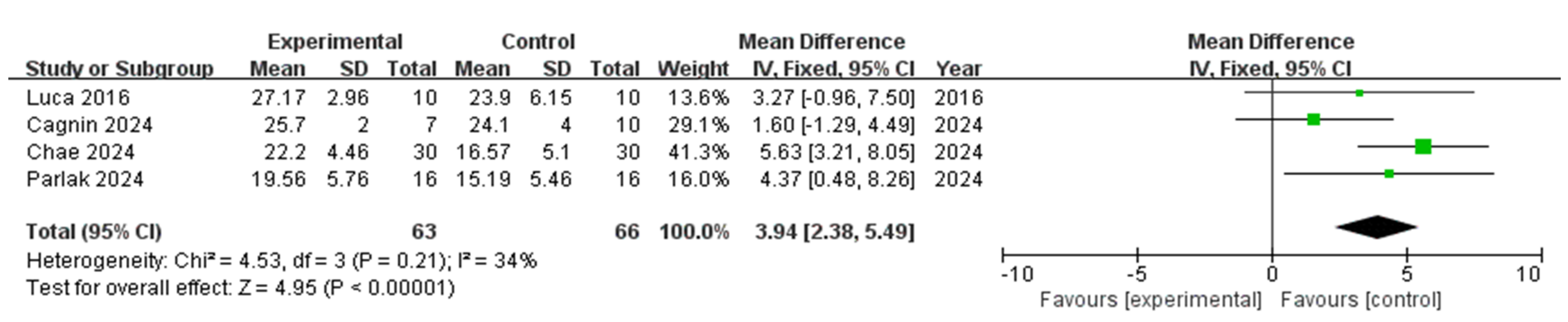

Figure 3-1. Effect of digital cognitive training interventions for dementia on MMSE(Mini-Mental State Examination)

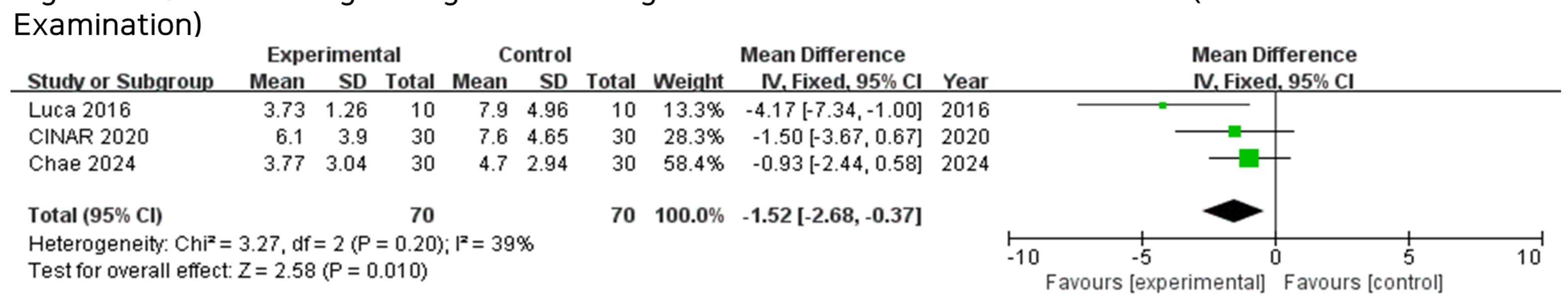

Figure 3-2. Effect of digital cognitive training interventions for dementia on GDS(Geriatric Depression Scale)

- The meta-analysis was conducted on studies that provided analyzable data on the major assessment instruments, MMSE(4 studies) and GDS(3 studies).
- The meta-analysis revealed that the experimental group significantly **improved MMSE Index** (Mean Difference: 3.94, 95% CI: 2.38 to 5.49, P < 0.00001) and **reduced GDS Index** (Mean Difference: -1.52, 95% CI: -2.68 to -0.37, P = 0.01).

CONCLUSIONS

- The meta-analysis demonstrates that the studies reviewed **support the effectiveness** of digital cognitive training interventions in dementia care.
- Digital cognitive training interventions have **the potential** to enhance the quality of life for dementia patients, reduce the burden on families and healthcare providers, and improve the long-term effectiveness of dementia care.

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